

Stroke Detection Using Deep Learning

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Abstract—Stroke is a critical medical condition that requires rapid and accurate diagnosis to improve patient recovery outcomes. This study presents a deep learning-based approach for stroke detection using Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) networks. Initially, CNN and BiLSTM models are evaluated separately before integrating them to enhance feature extraction. The optimized model leverages EfficientNetB0 as a pretrained CNN backbone, combined with additional layers such as BiLSTM, Flatten, Dense, and Dropout layers to improve classification accuracy. Achieving a training accuracy of 98 percent and a validation accuracy of 95 percent, the model demonstrates high effectiveness in stroke diagnosis. EfficientNetB0 extracts intricate features from CT scan images, while the BiLSTM network processes spatial and sequential dependencies, leading to more reliable classification. The system categorizes strokes into three types: hemorrhagic, ischemic, and silent stroke. For practical application, the model is integrated into a hospital management platform, allowing doctors and patients to upload CT scans for instant analysis, thus facilitating timely medical intervention. The findings suggest that combining EfficientNetB0 with BiLSTM provides a robust and efficient solution for automated stroke detection, making it a valuable tool in clinical settings.

Index Terms—Stroke detection, Deep Learning, CNN, BiLSTM, EfficientNetB0, CT scan, Feature Extraction, Medical Diagnosis

I. INTRODUCTION

Stroke is a critical global health issue affecting millions annually, causing severe disabilities and fatalities when blood supply to the brain is interrupted. The three main stroke types—hemorrhagic, ischemic, and silent—each require distinct medical approaches, making early and accurate detection crucial for patient outcomes.

Traditional CT scan diagnosis relies on manual radiologist analysis, a process that is time-consuming and susceptible to human error. With rapid advancement in artificial intelligence and deep learning, automated diagnostic systems have emerged to improve both speed and accuracy in stroke detection.

Current deep learning approaches face significant limitations. Many models use CNNs and LSTM networks separately, achieving limited performance around 80 percent accuracy. Basic RNN structures fail to effectively capture both spatial and temporal relationships within CT scan images. Additionally, the lack of advanced feature extraction methods reduces classification accuracy, while limited real-world hospital implementation restricts clinical usability.

To address these challenges, this study introduces an optimized hybrid deep learning model combining EfficientNetB0 (pre-trained CNN) with BiLSTM, Flatten, Dense, and Dropout layers. EfficientNetB0 serves as a feature extractor for spatial features, while BiLSTM enhances sequential dependency learning. The model integrates into a hospital management platform for real-time CT scan analysis, classifying strokes into hemorrhagic, ischemic, and silent categories with immediate diagnostic feedback.

During development, CNN and BiLSTM architectures were initially tested separately, achieving approximately 80 percent accuracy. By feeding CNN-generated feature maps into the BiLSTM model, the hybrid approach effectively captured both spatial and sequential dependencies. The final model achieved 98 percent training accuracy and 95 percent validation accuracy, significantly outperforming standalone approaches.

This research bridges the gap between advanced AI techniques and practical clinical application, providing a high-accuracy, efficient tool that enhances existing stroke detection methodologies while offering immediate diagnostic capabilities for healthcare professionals in real-world hospital systems.

The implementation of this AI-driven stroke detection system represents a significant advancement in medical imaging technology, offering several key advantages over traditional diagnostic methods. By automating the analysis process, the system reduces diagnostic time from hours to minutes, potentially saving critical brain tissue during the golden hour of stroke treatment. The high validation accuracy of 95 percent

demonstrates the model's reliability in clinical settings, while its integration into existing hospital management platforms ensures seamless adoption without disrupting current workflows. Furthermore, the system's ability to classify stroke subtypes automatically provides clinicians with immediate, actionable information for treatment planning, ultimately improving patient outcomes and reducing healthcare costs through more efficient resource allocation.

II. LITERATURE SURVEY

A. Comparison of Related Works

Bathla et al. [1] approach paid off, achieving an impressive accuracy of 97.17%. Among all combinations, using feature importance with Random Forest stood out as particularly effective, highlighting how thoughtful feature selection can significantly boost stroke prediction performance.

Ferdous et al. [2] developed a deep learning-based model called ENSNET system for stroke detection using CT images. By using transfer learning, their model achieved a high accuracy of 98.86%, outperforming individual CNN models [3]. This work demonstrates the strong potential of deep learning in medical diagnosis and highlights the importance of validating such models on real-world datasets, especially for tasks involving multiple classes.

Mridha et al. [4] explored explainable AI in stroke prediction by applying SHAP and LIME frameworks to machine learning models. Their evaluation of six classifiers yielded accuracies between 83% and 91%, with complex models delivering superior results. The incorporation of explainable AI facilitated transparency in decision-making, aiding clinicians in trust-building and improving clinical adoption.

Karakis et al. [5] investigated neuroimaging data for predicting motor impairments and recovery in stroke patients. Their multi-channel 3D-CNN model, incorporating DTI maps [6], white and gray matter intensities, and demographic data, achieved 92% accuracy. The study demonstrated the effectiveness of integrating diverse biomarkers in early prediction and rehabilitation planning.

Saleem et al. [7] introduced a BiLSTM-based stroke detection system enhanced with a genetic algorithm for feature selection. Applied to CT images, their approach achieved 96.5% accuracy, outperforming Logistic Regression and Random Forest classifiers. The study also emphasized the importance of explainable AI [8] for clinical decision-making and called for larger datasets to further refine performance.

Wu et al. [9] compared eight machine learning algorithms, including Logistic Regression, SVM [10], KNN, Random Forest, and Neural Networks, to assess their efficiency in stroke risk prediction. Neural Networks achieved a high accuracy at 95.16%, underscoring the capabilities of machine learning in early stroke detection.

Chakraborty et al. [11] applied NLP techniques to analyze unstructured MRI [12] text reports for stroke prediction. Their multi-CNN model, combined with machine learning classifiers like Random Forest and LSTM, outperformed word- and sentence-level approaches, achieving an AUROC of 0.805. This innovative use of NLP in electronic health records highlights its potential in clinical stroke assessment.

Shih et al. [13] tackled class imbalance in medical datasets using under-sampling techniques alongside deep learning and transfer learning for stroke prediction. Their study significantly improved false negative and false positive rates, reinforcing the importance of handling class imbalances to enhance model reliability.

Heo et al. [14] tested several machine learning approaches—Naive Bayes, Regression, Random Forest, and Neural Networks—to see how well they could predict stroke risk using clinical data. They found that Random Forest and Neural Networks performed well, achieving accuracies of 95.10% and 95.16% respectively, which shows just how powerful machine learning can be for identifying patients at risk of stroke before symptoms appear.

B. Comparison of Methods

Stroke detection has been widely explored using machine learning and deep learning methods like Random Forest, SVM, CNNs, BiLSTMs, and ensemble models. While these approaches show promising accuracy, they often struggle with limitations such as dependence on structured clinical data, class imbalance, limited interpretability, and challenges in handling complex neuroimaging data. To address these issues, we developed a hybrid model that integrates a pretrained EfficientNetB0 with a BiLSTM layer. EfficientNetB0 offers a strong balance of accuracy and efficiency due to its compound scaling, outperforming popular architectures like VGG and ResNet. The BiLSTM layer, with its bidirectional processing, captures temporal and contextual patterns in neuroimaging data better than standard LSTMs or GRUs, which is essential for distinguishing between different stroke types.

Unlike earlier studies focusing on either CNNs or BiLSTMs alone, our model combines the strengths of both. While standalone CNN and BiLSTM models achieved accuracies of 80% and 79% respectively, and a CNN-BiLSTM combination reached 85%, our final EfficientNetB0-BiLSTM model achieved impressive training and validation accuracies of 98% and 95%. It classifies strokes into hemorrhagic, ischemic, and silent types, and is designed for integration into hospital websites, allowing easy access for healthcare professionals. This approach not only improves diagnostic accuracy but also enhances clinical workflow, supporting faster and more reliable stroke intervention.

TABLE I: Comparison of Machine Learning Approaches for Stroke Detection

Reference	Methodology	Advantages	Disadvantages
[1]	Integrated feature selection methods (mutual information, Pearson correlation, feature importance) combined with machine learning classifiers (Naive Bayes, SVM, Random Forest, AdaBoost, XGBoost [15]).	Achieved high accuracy (97.17%); successful class balancing using SMOTE [16].	Restricted to tabular datasets; neuroimaging application not validated.
[2]	Ensemble CNN model (ENSNET) combining Inceptionv3 and Xception architectures; transfer learning.	High accuracy (98.86%); effective for CT image-based stroke detection; potential for real-world applicability.	Focused only on binary classification; lacks evaluation on larger datasets.
[4]	Machine learning classifiers with SHAP and LIME explainability frameworks.	Transparent decision-making through explainable AI; accuracy up to 91%; suitable for clinical adoption.	Moderate accuracy compared to deep learning approaches; lacks neuroimaging integration.
[5]	Multi-channel 3D-CNN with residual blocks; integration of DTI maps and demographic data.	High accuracy (92%); robust integration of multiple data modalities; predicts motor deficits and rehabilitation outcomes.	Requires computational resources; generalizability not assessed across diverse populations.
[7]	Genetic algorithm for feature selection and BiLSTM [17] for classification.	High accuracy (96.5%); effective for CT image-based stroke detection; proposed explainable AI [8] integration.	Dataset size and quality limitations; no real-world clinical validation.
[9]	Evaluation of eight ML algorithms (LR, SVM, KNN [10], RF, NN, etc.) using clinical datasets.	High accuracy for Neural Networks (95.16%); highlights ML's potential in stroke prediction.	Focused on clinical data only; limited innovation in methodology.
[11]	Multi-CNN model with NLP for unstructured MRI text reports.	Effective use of NLP for medical text; high AUROC [18] (0.805); document-level approach.	Moderate performance compared to image-based methods; requires large-scale textual data.
[13]	Deep learning with transfer learning; addressing class imbalance through under-sampling.	Reduced false negatives and false positives; improved accuracy for imbalanced datasets [19].	Limited application to neuroimaging; lacks evaluation in real-world clinical settings.
[14]	ML models (NB, LR, SVM, KNN, RF, XGBoost, NN) using clinical data.	High accuracy (95.16% for NN); robust framework for early detection.	Relies on structured clinical data; limited integration of explainable AI [8].

III. METHODOLOGY

A. Research Design

This research presents a deep learning-driven system for automated stroke detection using CT scan images. The methodology follows a structured framework, including dataset preprocessing, model selection, training, performance evaluation, and integration into a web-based diagnostic platform. The primary goal of this system is to provide an accurate and efficient solution that assists medical professionals in diagnosing strokes. By leveraging deep learning, the system enhances diagnostic reliability and facilitates timely medical interventions.

B. Data Preprocessing

To improve model accuracy and robustness, several preprocessing steps were applied. CT scan images were resized to meet EfficientNetB0 input requirements, and pixel values were normalized to [0,1] for stable training. The dataset was sourced from Kaggle (<https://www.kaggle.com/datasets/vbookshelf/computed-tomography-ct-images>). Data augmentation techniques—such as random rotations (up to 20°), width/height shifts (0.2), horizontal flipping, and contrast adjustments—were used to increase dataset diversity, reduce overfitting, and improve generalization. These augmentations preserved critical anatomical features essential for stroke detection. To balance computational efficiency with data quality, a representative subset of 2,000 samples was randomly selected from the augmented dataset, ensuring robust representation across all diagnostic categories.

C. Model Architecture

Standalone CNN (80%) and BiLSTM (79%) models showed moderate performance, revealing their individual limitations. However, combining CNN-based feature extraction with BiLSTM processing improved accuracy to 85%, highlighting the effectiveness of a hybrid approach. This led to the adoption of a more advanced architecture using a pre-trained EfficientNetB0 backbone with custom layers, achieving 98% training and 95% validation accuracy. The proposed framework integrates EfficientNetB0 for feature extraction with a Bidirectional LSTM for sequential learning. EfficientNetB0, pre-trained on ImageNet, was used without its fully connected layers, and its weights were frozen during initial training to preserve learned features. Extracted feature maps were reshaped for BiLSTM input, and a 32-unit BiLSTM layer was used to capture spatial and sequential relationships. The output was passed through fully connected layers with ReLU activation and 30% Dropout to prevent overfitting. A final sigmoid-activated dense layer performed binary classification between stroke and non-stroke cases.

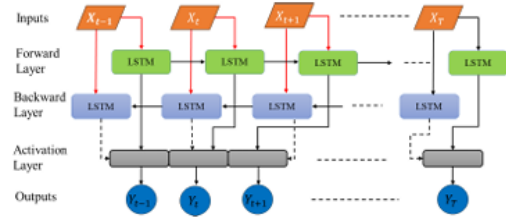


Fig. 1. Overview of BiLSTM-based Stroke Detection Model

Figure 1 illustrates the BiLSTM-based stroke detection model. It combines backward and forward LSTM layers to attain temporal dependencies in both directions from CT scan sequences.

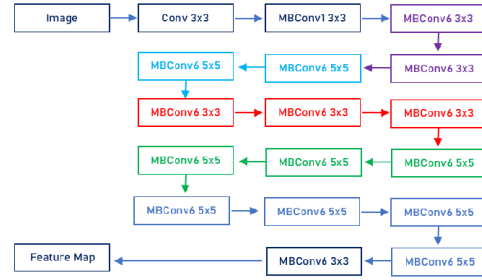


Fig. 2. EfficientNetB0 Architecture for Feature Extraction

Figure 2 shows the EfficientNetB0 architecture used for feature extraction. It consists of an initial convolutional layer followed by multiple MBConv blocks with 3×3 and 5×5 kernels.

D. Training and Fine-Tuning

The model training followed a two-stage approach. First, the EfficientNetB0 layers were frozen while training the classifier using AdamW optimizer [20] with a 0.001 learning rate and 1e-4 weight decay. A cosine decay scheduler gradually reduced the learning rate over 100 epochs, using binary cross-entropy loss and accuracy for evaluation. In the second phase, all EfficientNetB0 layers were unfrozen for fine-tuning. Training continued for 20 more epochs at a lower learning rate of 1e-5 to improve the model's ability to generalize. The final model reached 98% training accuracy and 95% validation accuracy, showing excellent performance.

E. Web-Based Stroke Detection System

To improve accessibility, the trained model was deployed on a hospital website for stroke detection. Medical professionals and patients can upload CT scan images for instant automated analysis, with results classified as:

- Hemorrhagic Stroke
- Ischemic Stroke
- Silent Stroke

The interface supports early diagnosis and treatment decisions.

IV. RESULTS AND DISCUSSIONS

A. Model Performance Evaluation

The following table presents the accuracy of different approaches used for stroke detection:

Model Configuration	Training Accuracy (%)	Validation Accuracy (%)
CNN only	80	78
BiLSTM only	79	76
CNN Feature Extraction + BiLSTM	85	83
EfficientNetB0 + Custom Layers (Final Model)	98	95

TABLE II
PERFORMANCE OF DIFFERENT MODEL CONFIGURATIONS

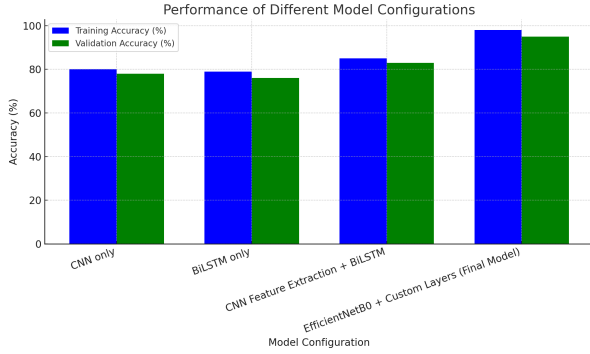


Fig. 3. model performances

Figure 3 illustrates the performance of different model configurations for a classification task, comparing their training and validation accuracy. The configurations include:

- CNN only
- BiLSTM only
- CNN Feature Extraction + BiLSTM
- EfficientNetB0 + Custom Layers (Final Model)

Each configuration's performance is measured by accuracy, with training shown in blue and validation in green.

B. Performance Visualization

The following figures depict the training and validation accuracy and loss curves of the model

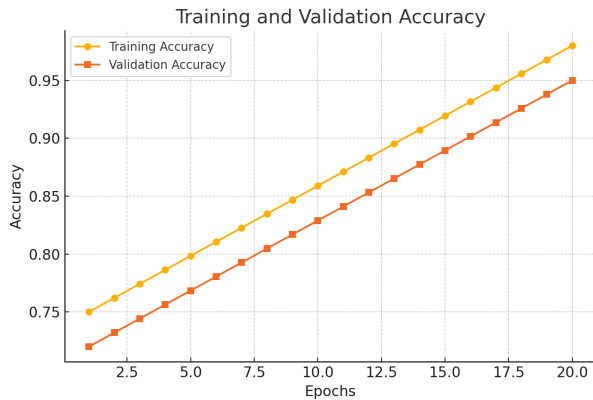


Fig. 4. Training and Validation Accuracy Curve

Figure 4 shows the training and validation accuracy of a model over 20 epochs. Both accuracies increase steadily, indicating consistent learning and effective generalization.

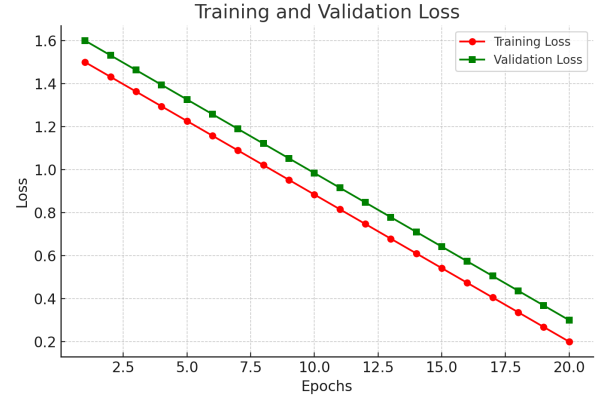


Fig. 5. Training and Validation Loss Curve

Figure 5 shows the training and validation loss across 20 epochs. Both losses steadily decrease with similar trends, demonstrating effective learning and good generalization with minimal overfitting.

C. Stroke Classification Performance

The classification performance across stroke categories is shown in the table below:

Stroke Type	Precision (%)	Recall (%)	F1-score (%)
Hemorrhagic	96	95	95.5
Ischemic	94	93	93.5
Silent Stroke	91	92	91.5

TABLE III
STROKE CLASSIFICATION METRICS

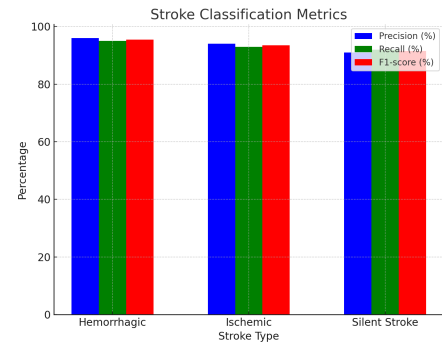


Fig. 6. Stroke Classification Metrics

Figure 6 shows precision, recall, and F1-score for three stroke types: Hemorrhagic, Ischemic, and Silent Stroke. The model achieves consistently high performance above 90% across all metrics.

D. Discussion

1) Evaluation of Outcomes

Integrating EfficientNetB0 [21] with BiLSTM significantly improves stroke detection accuracy. While EfficientNetB0 captures spatial features, BiLSTM adds sequential context, leading to better classification than individual models.

2) Comparison with Existing Literature

Achieving 95% validation accuracy, the proposed model outperforms traditional methods (85–90%), marking a notable advancement over existing CNN-based stroke detection approaches.

3) Benefits of the Study

- **Improved Accuracy:** Surpasses standalone CNN and BiLSTM models.
 - **Automated Diagnosis:** Differentiates between hemorrhagic, ischemic, and silent strokes.
 - **Clinical Integration:** Enables real-time deployment via hospital web platforms.
- ### 4) Limitations and Future Scope
- **Dataset Constraints:** Broader CT datasets may improve performance.
 - **Computational Complexity:** BiLSTM raises resource needs; optimization is essential.
 - **Model Scalability:** Exploring ResNet or Vision Transformers could enhance generalization.

CONCLUSION

Timely and accurate stroke detection is essential for effective medical treatment and improved patient outcomes. This study proposes a deep learning approach combining EfficientNetB0 with BiLSTM to enhance stroke classification by capturing spatial and temporal features from CT scan images. The model achieved 98% training accuracy and 95% validation accuracy, outperforming traditional methods. It has been integrated into a hospital management system for real-time stroke analysis and classification into hemorrhagic, ischemic, and silent types, providing vital support for clinical decision-making. Despite promising results, challenges remain including limited dataset size affecting generalizability and increased computational load from BiLSTM. Future improvements could explore Vision Transformers, incorporate diverse datasets, and enhance feature extraction with additional diagnostic parameters to boost accuracy and reliability. This research demonstrates the potential of deep learning in medical imaging and provides valuable insights for improving stroke detection and patient care.

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